

Problem with training your own model

- 1- Data Hungry
- 2- Lots of time required

Transfer learning

Transfer learning is one of most important design technique of deep learning that works with images or objects recognition.

During the process of transfer learning, we need to decide

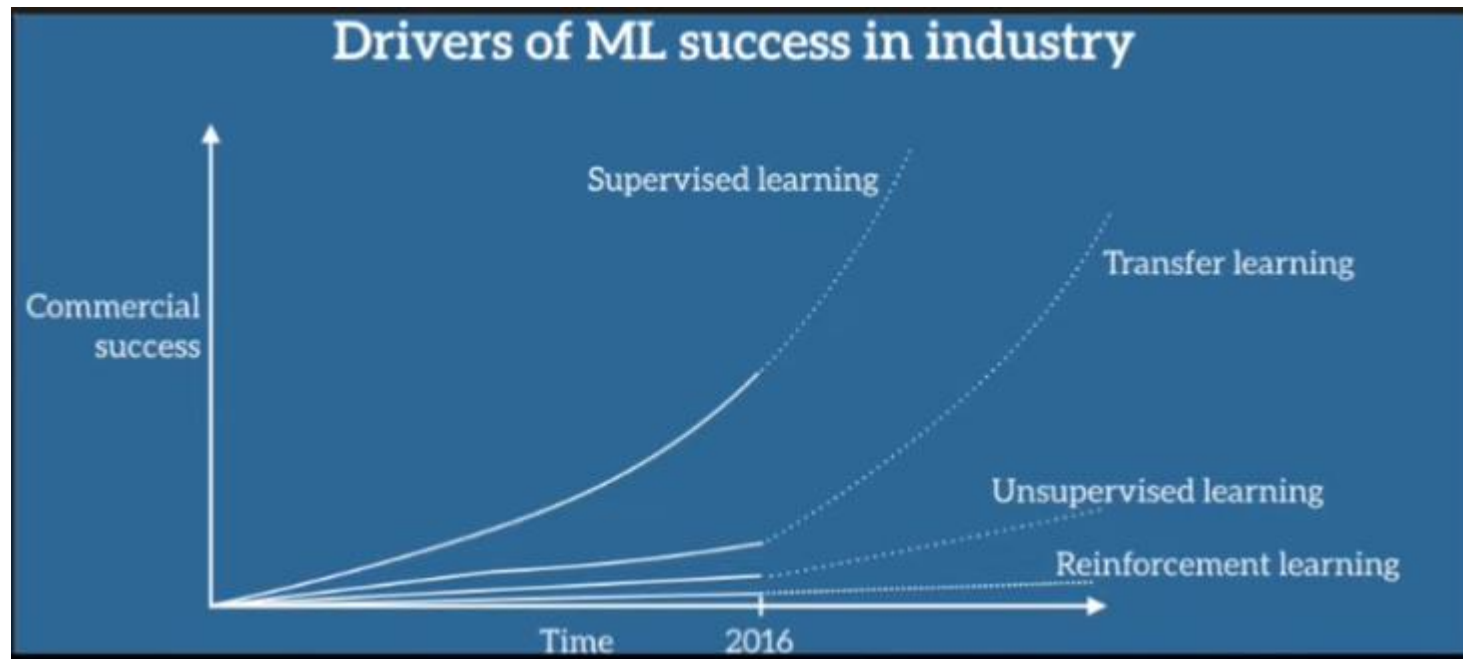
1. Which part of the knowledge can be transferred from the source to the target models in order to improve the performance of the target task?

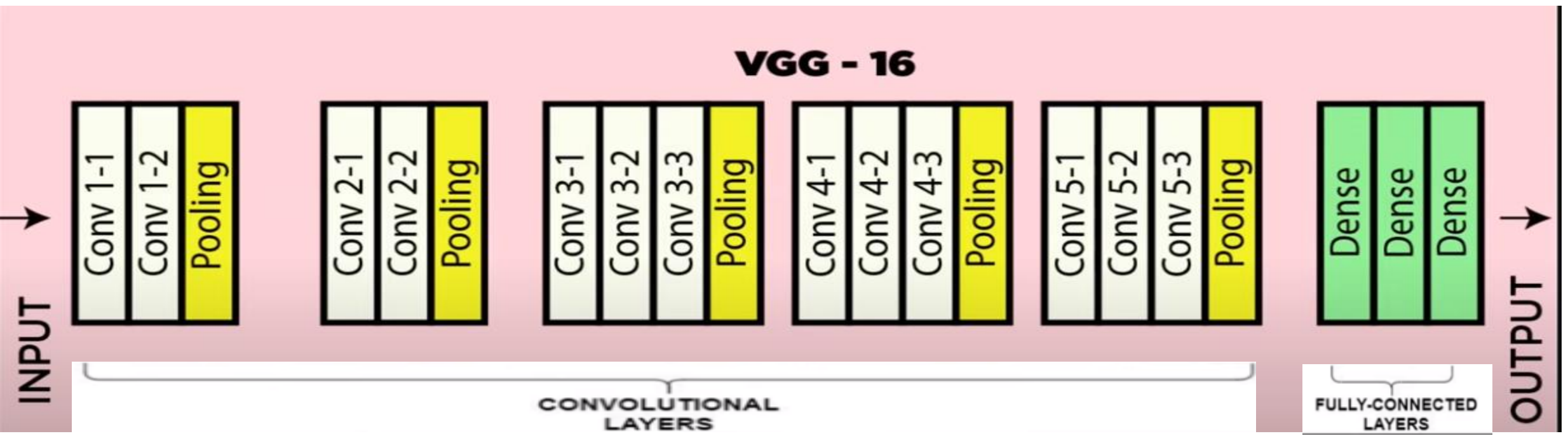
For this we try to identify what is common between the source and target model.

2. Aim of transfer learning is to improve target task performance results and not to degrade them. To achieve this, we need to be careful about where to transfer the knowledge and when not to.

Transfer learning Cont...

Transfer learning is a research problem in machine learning that focuses on storing knowledge gained while solving one problem and applying it to a different but related problem.





General features Like:
Edges, textures, lines,
shapes, colours

High Level features like:

Object parts: Eyes, nose, wings and etc

Pattern : like stripes on a zebra

Scenes : Contextual information such as beach, city or forest

Objects : Specific objects like cars, trees or building

Ways of doing Transfer Learning

1. Feature extraction
2. Fine Tuning

Imaget Net dataset

<https://gist.github.com/yrevar/942d3a0ac09ec9e5eb3a>

Feature Extraction

```
[4] import tensorflow
    from tensorflow import keras
    from keras import Sequential
    from keras.layers import Dense, Flatten
    from keras.applications.vgg16 import VGG16
```

```
model = Sequential()
model.add(conv_base)
model.add(Flatten())
model.add(Dense(256, activation='relu'))
model.add(Dense(1, activation='sigmoid'))
```

```
conv_base = VGG16(
    weights='imagenet',
    include_top=False,
    input_shape=(150, 150, 3))
```

```
[ ] model.summary()
```

```
Model: "sequential"
_____
Layer (type)                Output Shape              Param #
=====
vgg16 (Functional)          (None, 4, 4, 512)        14714688
flatten (Flatten)           (None, 8192)              0
dense (Dense)                (None, 256)              2097408
dense_1 (Dense)              (None, 1)                 257
=====
Total params: 16,812,353
Trainable params: 16,812,353
Non-trainable params: 0
```

```
conv_base.trainable = False
```

```
batch_size = 32
```

```
train_datagen = ImageDataGenerator(  
    rescale=1./255,  
    shear_range=0.2,  
    zoom_range=0.2,  
    horizontal_flip=True)
```

```
test_datagen = ImageDataGenerator(rescale=1./255)
```

```
train_generator = train_datagen.flow_from_directory(  
    '/content/train',  
    target_size=(150, 150),  
    batch_size=batch_size,  
    class_mode='binary')
```

```
validation_generator = test_datagen.flow_from_directory(  
    '/content/test',  
    target_size=(150, 150),  
    batch_size=batch_size,  
    class_mode='binary')
```

```
history = model.fit_generator(  
    train_generator,  
    epochs=10,  
    validation_data=validation_generator)
```

Fine tuning

```
conv_base.trainable = True

set_trainable = False

for layer in conv_base.layers:
    if layer.name == 'block5_conv1':
        set_trainable = True
    if set_trainable:
        layer.trainable = True
    else:
        layer.trainable = False

for layer in conv_base.layers:
    print(layer.name, layer.trainable)
```

Summary

```
input_1 False
block1_conv1 False
block1_conv2 False
block1_pool False
block2_conv1 False
block2_conv2 False
block2_pool False
block3_conv1 False
block3_conv2 False
block3_conv3 False
block3_pool False
block4_conv1 False
block4_conv2 False
block4_conv3 False
block4_pool False
block5_conv1 True
block5_conv2 True
block5_conv3 True
block5_pool True
```

Article

https://lisaong.github.io/mldds-courseware/03_TextImage/transfer-learning.slides.html